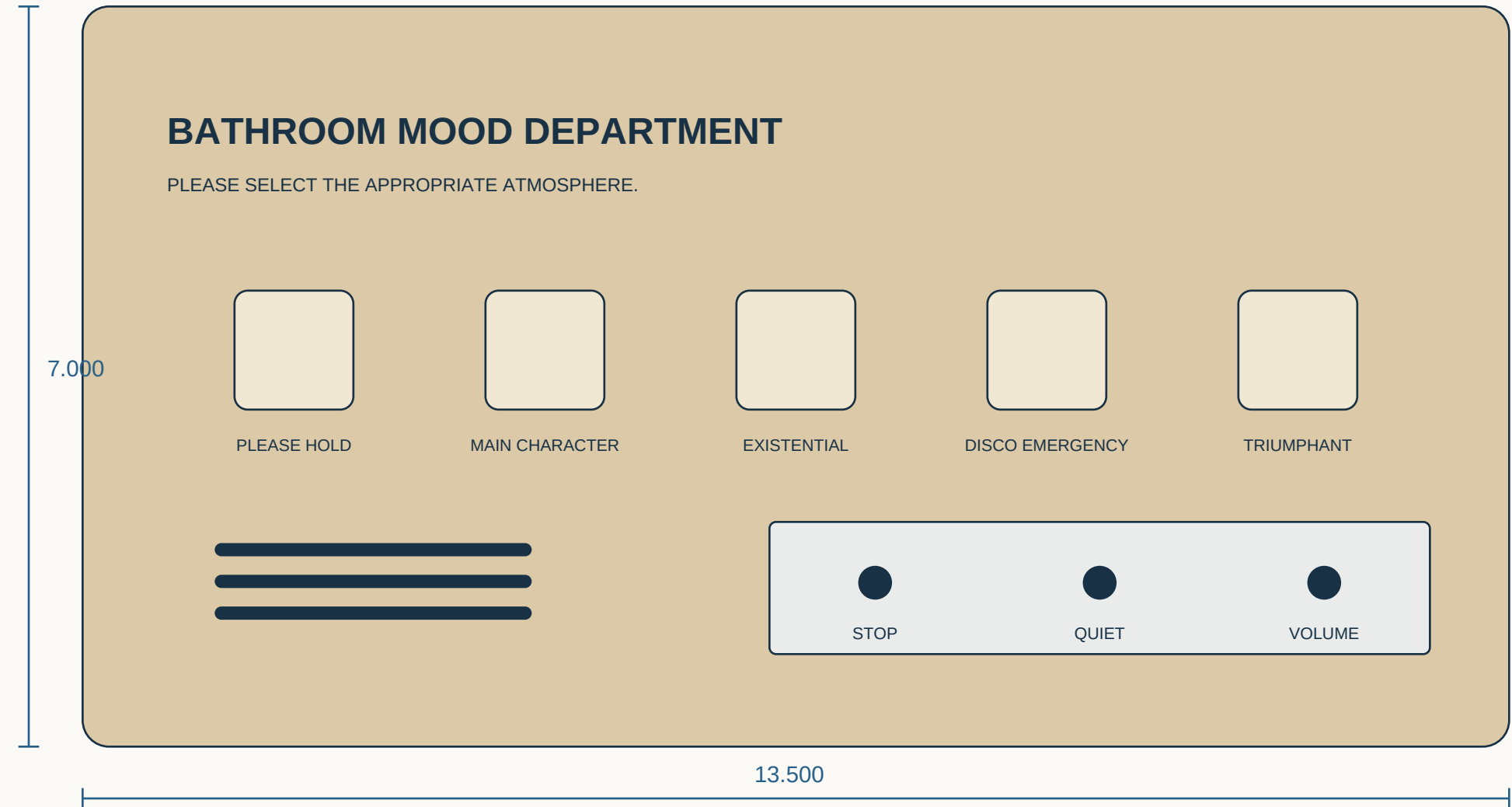




FIRST BUILD

A small musical mood selector with five wood buttons. Dry bench first; then a dry location outside the bathroom.

Start with one scrap button coupon and a separate sound test. Those two tests resolve the main uncertainty before the full walnut enclosure.

Five moods: Please Hold, Main Character, Existential, Disco Emergency and Triumphant. Use original or properly licensed short audio.

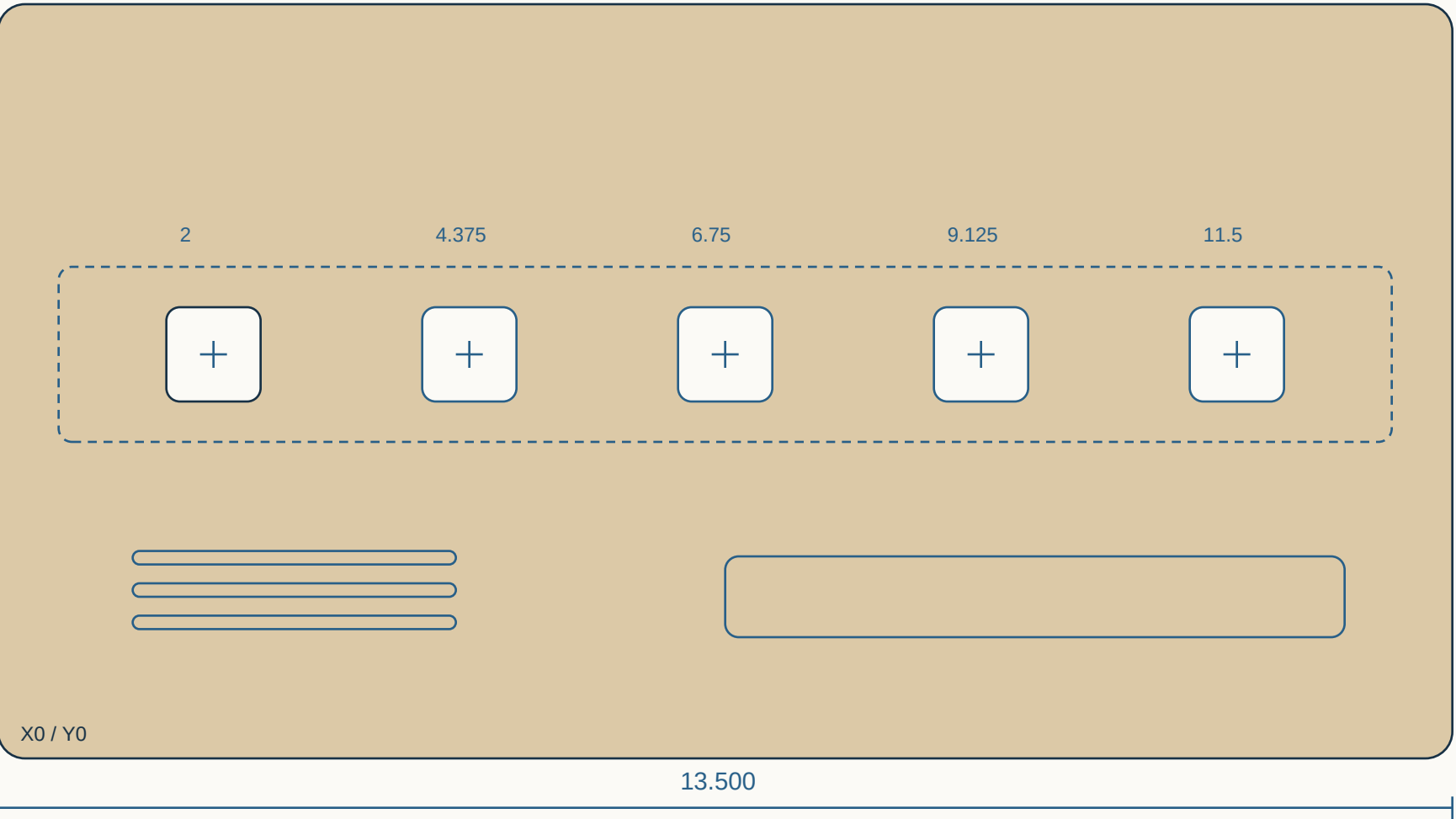
Physical QUIET, a STOP button and volume control. No microphone, wireless link or battery in R01.

Pico 2 + DFPlayer Mini + enclosed speaker. Reuse this basic arrangement in later sound-driven products.

FINISHED BODY

13.500 W x 7.000 H x 2.000 D; excludes button protrusion, controls, cable and wall cleat.

Appearance sketch: layout and materials proposed. Existing Quite Sharp branding remains in place.



FACE / FRONT PROJECTION

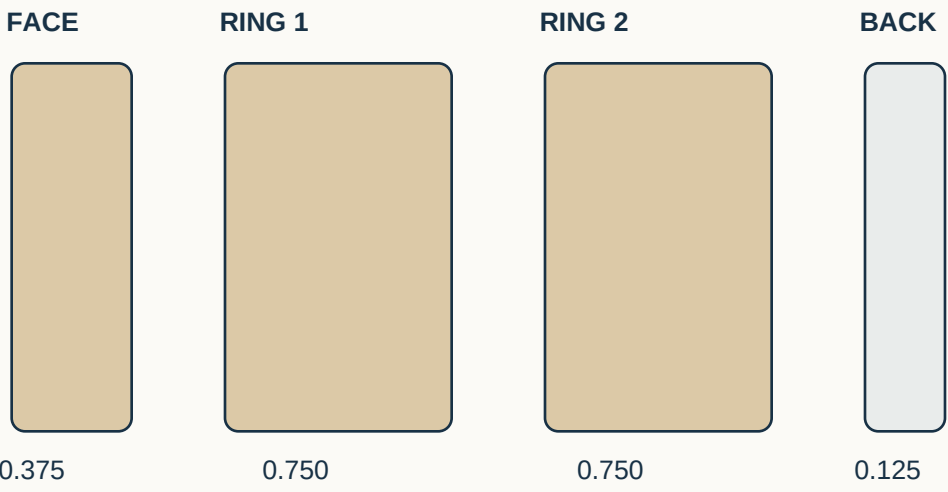
Face: 0.375 thick. All five opening centres Y3.750. X coordinates are shown above holes. Openings 0.875 square, corner R0.125.

Dashed rear recess: X0.5625 / Y2.9375, 12.375 x 1.625, R0.125. Depth 0.250 FROM REAR; leaves 0.125 skin. Reverse setup requires coordinate transform.

Grille: three 3.000 x 0.125 slots, end R0.0625. Lower-left X1.250; Y1.200, 1.500, 1.800.

Control opening: X6.750 / Y1.125; 5.750 x 0.750, R0.125. Metal plate and controls mount separately.

BODY STACK / NOT TO SCALE

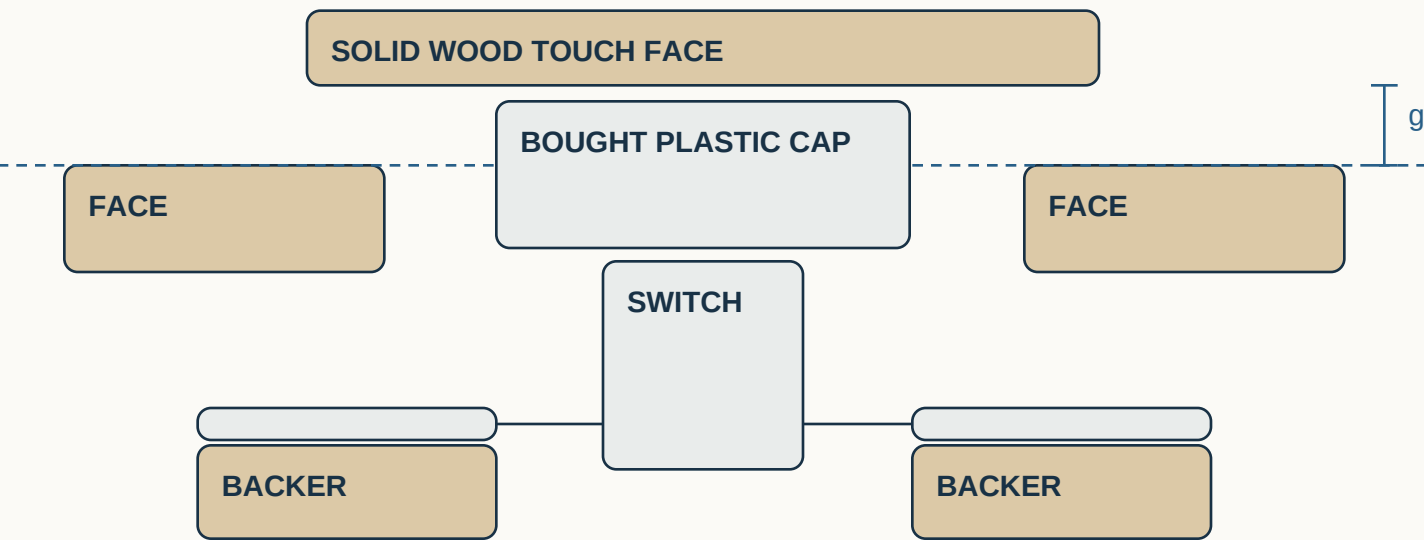


Two frame rings: 13.500 x 7.000 x 0.750 each. Inner opening X0.500 / Y0.500, 12.500 x 6.000, R0.125. Minimum rim 0.500. Grain along X in all three solid wood layers.

Removable 0.125 plywood rear. Eight 0.15625 x 0.3125 vertical screw slots. Centre screws and use washers; allow seasonal movement. Cleat screws engage frame, never thin back alone.

Outside corner R0.250. Prototype profile tolerance target +/-0.010; bought-part fits are holds. Stock allowance, cutters, tabbing and safe workholding are operator decisions.

SECTION THROUGH ONE KEY / SCHEMATIC, NOT SCALE



Carrier position adjusts with cleats / shims.
Wood overlap provides travel stop against face.

CONTROLLED DIMENSIONS

Wood face: 1.125 x 1.125 x 0.125; R0.125 corners. Panel opening: 0.875 x 0.875; R0.125.
Straight-side overlap: 0.125 each.

$g = 0.100$ inch target rest gap below wood face; trial range 0.095-0.110. Positive stop must occur AFTER electrical actuation and BEFORE switch bottom-out.

Measure H from carrier front to wooden cap top at rest. Carrier setback $P = H - 0.225$. Require $P \geq 0.145$ so plate clears the remaining face skin. P is NOT fixed until the coupon is measured.

Selected Kailh 4954: vendor travel 1.8 +/-0.3 mm to actuation; 3.6 +/-0.3 mm total. Trial carrier: 1.5 mm aluminium, 14 mm square opening. Verify clip engagement; file internal corners if needed.

COUPON BEFORE THE COMPLETE SIGN

Scrap face 3 x 3 x 0.375. Centred 0.875 opening. Rear recess 2.5 x 2.5 x 0.250, leaving 0.125 skin. Separate 2 x 2 aluminium carrier. Both DXFs included.

Keep plastic stem connection intact. Bond the wood overlay to a spare compatible cap, using adhesive confirmed for its actual plastic. Cure off-switch. Bought cap must clear panel opening through its full travel.

Test at least 100 centre and corner presses, then after finishing. No sticking, loose bond, false repeats or switch bottom-out. Record H/P and actual travel. This is a screening test, not a lifetime rating.

CARRIER / FULL PANEL

12.250 x 1.500 aluminium. Global lower-left X0.625/Y3.000. Local key X = 1.375, 3.750, 6.125, 8.500, 10.875; all Y0.750. Apertures 14 mm nominal, tagged HOLD_MX_FIT.

Add 0.500-thick wood backer, same outline, five 0.875 square through clearances. Four end fixings pass through both into cleats. Cleat front is behind backer at $Z=P+0.559$; see guide for four between-key plate screws.

Test complete row for flex: trial target ≤ 0.005 inch at normal actuation force, with all keys reliable. Coupon alone cannot validate span stiffness. See guide for clearances and fastener schedule.

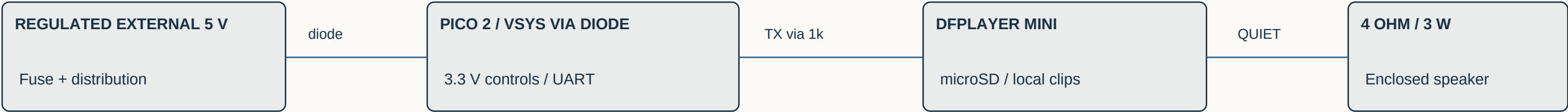


Diagram is functional: separate +5 V branch feeds DFPlayer directly. Common logic/power GND. Speaker is NOT grounded.

PIN MAP / PICO LABEL -> PHYSICAL PIN

UART TX	GP0 -> 1	1k series to DFPlayer RX
Mood 1-5	GP2..6 -> 4,5,6,7,9	10k pull-up; switch to GND
STOP	GP7 -> 10	10k pull-up; momentary to GND
QUIET sense	GP8 -> 11	10k pull-up; SOUND throw to GND
Volume	GP26 -> 31	10k pot on 3V3; filtered wiper
Supply	VSYS -> 39 / GND -> 38	External diode; VBUS40 unused

Physical QUIET opens one speaker lead with one DPDT pole; its other pole reports state. Never tie SPK1 or SPK2 to ground. DFPlayer TX and BUSY stay unconnected. See electrical review for exact wiring.

Firmware: debounce, no repeat on hold, latest selection wins, STOP/QUIET cancel pending play, no boot sound. TX-only cannot confirm playback or missing card. Initial ceiling 15/30, begin testing near 5.

BUILD IN THIS ORDER

1. Receive samples; make coupon and measure stack.
2. Bench wire; test inputs first with audio disconnected.
3. Program; add original/licensed files on microSD.
4. Test low-volume audio, STOP, QUIET and power-up.
5. Machine body only after coupon and audio pass.
6. Cure finish; mount serviceable hardware and wiring.
7. Verify back, cable clearance and rated wall cleat.

Power adapter voltage tolerance and Canadian approval marks must be checked on the actual unit. No internal mains, batteries, wet-location or sale certification is provided by this prototype.

Programming: remove external power AND audio harness, including TX, before connecting USB. Never let an unpowered player be fed through its signal wire.